

Harpreet Khaira

Senior AI Engineer | Production Agents, Orchestration, Evals & Backend Systems

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TECHNICAL SKILLS

AI & Agents: Multi-Agent Systems, Agent Orchestration, Agent Memory, Multistep Reasoning, Tool Use, MCP, RAG

Reliability: LLM Evals, LLM-as-Judge, Observability, Tracing, Telemetry, Verification Loops

Backend & Data: Python, TypeScript, FastAPI, R, Rust, Kafka, Neo4j, BigQuery, Elasticsearch, PostgreSQL

Cloud & Delivery: AWS, GCP, Docker, CI/CD, Distributed Systems, MLOps, A/B Testing

PROFESSIONAL EXPERIENCE

Senior AI Engineer | *BioRender* Jun 2025 – Jul 2026
0-to-1 agentic scientific analysis product

- Architected and productionized a **CrewAI-based multi-agent system** that transformed raw experimental data into validated analyses, visualizations, and conclusions, accelerating manual analysis cycles by up to **10x**.
- Built hybrid deterministic and agentic pipelines for **Excel ingestion, schema inference, data normalization, analytical-table generation**, statistical computation, and research-thesis validation.
- Developed the agent infrastructure end to end, including **orchestration, memory, tool integrations, MCP servers, multistep reasoning**, and domain-specific analysis capabilities.
- Implemented **Datadog telemetry, tracing, logging, and hybrid evaluations**, combining programmatic checks with LLM-as-judge rubrics to diagnose failures and measure analysis quality.
- Owned architecture and productionization across **Python, FastAPI, R, and product-facing services**; separated agent judgment from deterministic execution to keep results reproducible and auditable.
- Founded and led the **AI Guild for 10–15 engineers**, establishing standards for agent design, specifications, verification loops, and AI-assisted development.

Senior Software Engineer / Technical Lead | *Target* Oct 2022 – May 2025
Enterprise data platform & Graph RAG

- Built a production data platform synthesizing status and dependency signals from **100+ sources** for enterprise Networking, Security, and Change Management teams.
- Implemented a **Graph RAG context-discovery system**, increasing daily active usage by **10x**; built pipelines processing **terabytes of daily data**.
- Scaled the team from **2 to 5 engineers** as technical lead, owning architecture, delivery quality, mentoring, and cross-team execution.

Senior Software Engineer | *Autodesk* Feb 2018 – Oct 2022
Autodesk Drive – document lifecycle at scale

- Led backend development for **Autodesk Drive**, serving **100K+ MAUs**; scaled services to **50K+ req/min** and improved customer retention by **200%**.
- Built CI/CD and experimentation workflows that increased shipment rate by **10x**; production monitoring reduced mean time to detect by **300%**.

Software Engineer | *Bank of America* Jul 2014 – Jun 2016
Built an award-winning cross-browser automation framework that reduced testing effort by **96%**.

SELECTED PROJECTS

PaperPulse AI – Research-paper summarization and reproduction assistant; reduced replication time by 75%. [GitHub](#) **RL Policy Optimization** – Value iteration, Q-learning, and Monte Carlo methods; reduced simulated crash rate by 90%.

EDUCATION

Carnegie Mellon University – M.S. Computer Software Engineering 2016 – 2017
Vellore Institute of Technology – B.Tech Computer Science 2010 – 2014
Stanford University – AI Certificate: Deep Learning, Reinforcement Learning, AI Ethics 2024